① Aspect 4 (Highest Relevance)

Read, decode, and make sense of statements to create a model of the situation

apartment-purchase-qs1 typically presents a **realistic and complex apartment-purchasing scenario**, such as:

- Apartment price
- Down payment percentage
- Loan amount or additional costs (e.g. taxes, fees)

Students must first:

- Accurately interpret the written information
- Distinguish essential data from background details
- Construct a coherent mental model of the "purchase → payment → constraints" situation

Thus, the most central *Formulate* competency assessed is **constructing a situational model from real-world text**.

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② Aspect 6

Identify and describe the mathematical aspects of a real-world problem

Students must extract mathematical elements from the purchasing context, including:

- Total apartment price
- Percentages (down payment rates, taxes, etc.)
- Monetary amounts (loan size, upfront payment)

The task requires students to recognise that:

- The situation involves **relationships between totals and percentages**
- The real-world context has an underlying mathematical structure

③ Aspect 2

Identify the key variables in a model

Before any formal modelling or calculation, students must clearly identify key variables, such as:

- Total price of the apartment
- Down payment amount
- Loan amount
- Additional fees or constraints

Clear identification of variables is essential for subsequent modelling and problem solving.

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④ Aspect 1 / Aspect 3

Select a mathematical description or an appropriate representation

Although information is often given in words or tables, students still need to:

- Choose appropriate mathematical expressions (e.g. *down payment = total price × percentage*)
- Decide whether to represent information using monetary values, percentages, or a combination of both

This reflects the ability to **select suitable mathematical representations** within a real-world context.

⑤ Aspect 12

Identify assumptions, constraints, and simplifications in a model

The problem implicitly includes several assumptions and constraints, such as:

- A fixed apartment price
- Constant percentages throughout the calculation
- Ignoring market fluctuations, interest rate changes, or individual financial differences

More advanced understanding recognises that:

- This is a **simplified purchasing model** valid only under the stated conditions.

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⑥ Aspect 7

Simplify or decompose a situation

To make the situation mathematically manageable, students typically:

- Break the purchasing process into stages (calculate down payment → calculate loan → check conditions)
- Transform a complex real-life decision into a sequence of solvable mathematical sub-problems

This demonstrates the *Formulate* skill of **decomposing a complex situation into manageable components**.

① Aspect 4（相关性最高）

Read, decode and make sense of statements … to create a model of the situation

📌 *apartment-purchase-qs1* 通常描述一个**真实而复杂的购房情境**，例如：

- 公寓价格
- 首付比例
- 贷款金额或其他费用（如税费、手续费等）

学生首先必须：

- 准确理解文字描述和给定信息
- 区分哪些信息是关键、哪些是背景
- 在头脑中形成一个“购房—付款—约束条件”的整体情境模型

👉 因此，该题最核心的 formulate 能力是 **从现实文本中构建情境模型。**

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② Aspect 6

Identify and describe the mathematical aspects of a real-world problem

📌 学生需要从购房情境中提取数学要素，例如：

- 房价（总价）
- 百分比（首付比例、税率等）
- 金额（贷款额、实际支付金额）

👉 题目要求学生意识到：

- 这是一个涉及 **比例、百分比与总量关系** 的问题
- 现实情境已内含明确的数学结构

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③ Aspect 2

Identify the key variables in a model

📌 在进一步分析前，学生必须明确模型中的关键变量，例如：

- 公寓总价
- 首付金额
- 贷款金额
- 可能的费用或限制条件

👉 对变量的清晰识别是后续建模和计算的前提。

④ Aspect 1 / Aspect 3

Select a mathematical description or an appropriate representation

📌 虽然题目通常以文字或表格给出，但学生仍需要：

- 选择合适的数学方式来表达情境（如用公式表示"首付 = 总价 × 比例"）
- 决定使用金额、比例还是两者结合来刻画问题

👉 这体现了在现实语境中 **选择合适数学表示** 的能力。

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⑤ Aspect 12

Identify assumptions, constraints, and simplifications in a model

📌 题目往往隐含若干假设或约束，例如：

- 房价固定不变
- 百分比在整个计算过程中保持不变
- 忽略市场波动、利率变化或个人差异

👉 高水平理解会意识到：

- 这是一个**简化的购房模型**，只在给定条件下成立。

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⑥ Aspect 7

Simplify or decompose a situation

📌 为了便于建模，学生通常会：

- 将购房过程拆分为几个步骤（计算首付 → 计算贷款 → 判断是否满足条件）
- 把复杂的现实决策问题转化为一系列可处理的数学子问题

👉 这体现了通过**分解复杂情境来实现数学建模**的 formulate 能力。